

A Peripheral Tactile Feedback System for Lateral Epicondylitis Rehabilitation Exercise

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ABSTRACT

Lateral epicondylitis (LE), or tennis elbow, is a highly prevalent musculoskeletal condition that affects millions of people. Physiotherapy is a common treatment, with a large portion consisting of prescribed home-based exercises. Adherence to these programs is an important factor in rehabilitation, however there are many barriers to adherence including the exercise taking up too much of the patient's attention, or the patient feeling like they are not carrying out exercises correctly. To address these problems, this paper describes a prototype system that uses haptic feedback to guide the patient to correctly carry out a commonly prescribed LE rehabilitation exercise, while allowing them to attend to external information such as another person or a screen. The system peripherally conveys information about the speed of movement and position of the user's wrist movement via peripheral vibration feedback, allowing the user to make adjustments to movement whilst keeping the visual and auditory senses free to attend to other sources. Finally, we discuss future areas of research for this prototype and applications of vibrotactile feedback for physiotherapy in general.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

Physiotherapy; Tennis Elbow; Haptic Interface; Vibrotactile; Parameter Mapping

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1 INTRODUCTION

Lateral epicondylitis (LE), lateral elbow tendinopathy, or most commonly *tennis elbow*, is a painful and potentially debilitating condition brought on by overuse of tendons in the forearm. It can cause physiological pain [14] and, in some cases, anxiety and depression [1]. The condition is highly prevalent, ranging from 1% to 3% of the adult population [26], with an estimated cost to the economy of £27 million per year in the UK alone due to required time away from work [11]. There are a number of treatment strategies including medication and steroid injections, but the least invasive and most accessible intervention is physiotherapy. Prescription of exercise programs to carry out at home are a common component of physiotherapy and a significant and prevalent hindrance for successful rehabilitation is poor adherence or complete abandonment of these programs [3]. Some of the barriers that have been shown to impede adherence include: low self-efficacy (belief in ability to initiate or carry out exercises correctly) [12]; some patients' belief that they would be more willing to carry out exercises if they were accompanied by a friend or partner [5]; and inconvenience or conflict with everyday life [5]. Further to this, the effects of the COVID-19 pandemic have meant that many non-emergency elective treatments (including physiotherapy [17]) have been postponed or services have been reduced to emergency-only cases. For conditions such as tennis elbow, where almost all cases are non-emergency, this means that treatment of these conditions has not only postponed, but when services resume they will understandably be secondary to more high-risk conditions.

This has meant that physiotherapy departments have looked to alternatives to providing treatment. One popular approach has been 'telerehabilitation', which is consulting via video or phone call [27]. However, access to these consultations is still challenging for many people, due to increased waiting times caused by over-stretched health services in addition to a growing backlog of elective treatments [17]. The combined effects of: the prevalence of LE [26], the quality of life impact that LE can have on sufferers [1, 14], the challenges posed to physiotherapy departments by the COVID-19 pandemic [17] and the economic impact caused by LE [11] mean that developing interventions that can: improve adherence to rehabilitation programs, improve accessibility to physiotherapy or reduce strain on physiotherapy departments are vitally important to mitigate these challenges.

In this paper, we present HALA (Haptically Augmented Lateral Elbow Tendinopathy Aid), a prototype system which uses vibration feedback to convey information to the user regarding the speed of movement and position of the wrist gathered from a sensor worn on the hand during *eccentric exercise of the wrist extensors*, which is

a commonly prescribed exercise for LE rehabilitation [4] (Figure 1). The aim being that through this feedback, a user can understand in real-time how well they are carrying out each repetition and can adjust their motion as required, in a similar fashion to the feedback a patient would usually receive from their physiotherapist. The system uses non-visual feedback, allowing the user to attend to some other task during exercise, thus confronting the barriers to adherence of: being able to be accompanied by a friend or partner during physiotherapy [5] and fitting physiotherapy around everyday life activities [5] such as conversations or watching television. Furthermore, this system aims to be a step towards developing technologies that can support physiotherapists during telerehabilitation to ensure that patients are performing exercises correctly. We hope that the prototype we propose can incite discussion in the community about how we can utilise HCI expertise in areas such as haptics to aide physiotherapists in reducing patients' barriers to adherence.

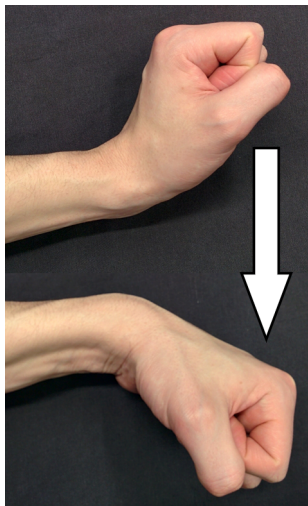


Figure 1: Eccentric contraction of the wrist extensors: the wrist is extended (raised) and flexed (lowered), with the forearm pronated (facing downward) at all times.

2 BACKGROUND

There have been a number of attempts in the human-computer interaction community to provide tools to assist users that are carrying out physiotherapy exercises at home. Ayoade & Baillie [2] used kinematic sensors to record and visualise the angle of the knee during exercises and a study with patients recovering from knee replacement surgery showed that patients using the system had higher rehabilitation outcomes than those who did not. The PhysioHome system used real-time visual guides projected on top of a mirror image of the patient to provide corrective feedback during home based physiotherapeutic exercise [25]. These systems all rely on visual feedback, however there has been research into the use of non-visual feedback in similar applications. Vibration based systems have been used to aid physical rehabilitation of conditions such as balance loss [30] and joint replacement [16]. Georgiou *et al.*

used vibrotactile feedback to guide patients with Huntington Disease during gait physiotherapy [9], they also noted the benefits to the patient of having the visual and auditory sense free for conversation etc. during physiotherapy. Scholz *et al.* applied *sonification* (the mapping of data to sound parameters) to convey stroke patients' body movements through sound [22] and found that patients whose movements were sonified during therapy showed improved motor skills compared to those who didn't use sonification. Singh *et al.* found that sonifying body movements of sufferers of chronic pain over two weeks resulted in increased self-efficacy [23].

These previous works exemplify the benefit of computer-based systems for aiding rehabilitation using a variety of feedback modalities, however only a small subset of this previous research focuses on home based rehabilitation [2, 23] and currently there are no such systems specifically targeting tennis elbow or similar conditions. The majority of the previously mentioned systems require a significant amount of user attention or even constant attention, such as the sonification systems that are constantly playing sound [22], resulting in reduced capacity to carry out any other tasks while exercising. This increases the probability of a number of the aforementioned barriers to adherence occurring. However, tactile feedback allows the visual and auditory senses to remain unencumbered, freeing attention for other audiovisual activities, such as watching TV or talking to others.

3 SYSTEM DESIGN

Motivated by this prior work, our goal is to develop a system which allows users to attend to other sources of information whilst correctly carrying out eccentric exercises of the wrist extensors. To correctly carry out this movement, the wrist must begin in an extended (upward bend) position and slowly flex downward with the full movement taking between 6 and 10 seconds [29]. The wrist is then carefully extended back to the starting position (Figure 1). In order to facilitate correct movement without imposing too much cognitive load on the user, HALA obtains data about the user's wrist movement and uses short vibrotactile cues to convey information about the movement including: *position of the wrist* and *speed of the movement* - indicating if the just-completed movement was too slow, too fast, or within the correct range - allowing the user to adjust timing on the next repetition. Finally, feedback is given when the correct number of repetitions has been reached to reduce the cognitive effort of keeping count. The hardware consists of an x-io Technologies NGIMU inertial measurement unit (IMU) worn around the hand which communicates the user's wrist movement data to a Unity application and this application provides vibration feedback to the user via a small Tactile Labs Haptuator MkII TL002-09-B actuator (Figure 2).

3.1 Feedback Design

The system needs to be used in the background, therefore the feedback must be as noninvasive as possible. The system is designed to confront this in two ways: using signals that are short (<3 secs) thus reducing the constant attention that continuous signals require; and using *parameter mappings*, which are the combination of data parameters (e.g. wrist position, movement speed) and the vibrotactile parameters used to represent them (e.g. duration), such that the

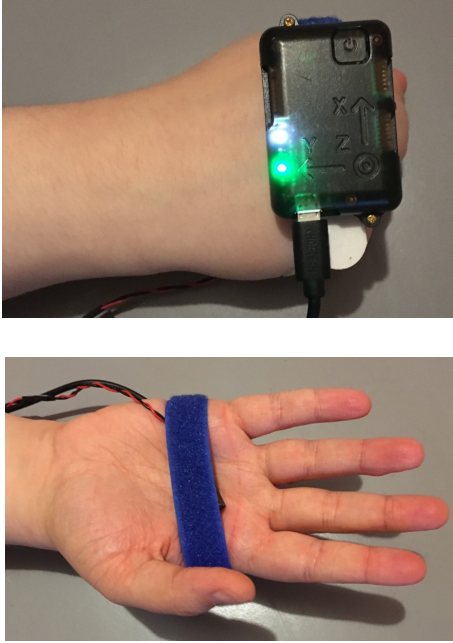


Figure 2: The hardware configuration for HALA: (upper) shows the IMU fitted around the top of the hand, with USB connection to the Unity application, (lower) shows the vibration actuator fitted on the palm side of the hand.

Data Parameter	Tactile Parameter	Mapping
Extension	Duration	Extended = 100ms Middle = 500ms Flexed = 1000ms
Speed	Pulse Tempo	Too Slow = 50/800ms In Range = Constant Too Fast = 50/15ms

All vibrations have a frequency of 200 Hz to ensure consistency with previous work [7].

Table 1: The feedback mapping design, showing the data parameter from the exercise to be conveyed, the tactile parameter that the data value is mapped to and the mapping for each cue.

relationship between the two is congruent for the user. Previous research suggests that tactile signals which are based on parameters that have generally negative connotations relating to the sense of touch, such as tactile roughness and rapid vibration pulses, are perceived as congruent with similarly semantically negative types of data, such as values relating to stress, error or danger, as well as more semantically neutral types of values like size and distance [7]. These studies provide polarity (the direction the relationship is perceived in) and scale (the ratio between the two parameters) data for a number of these data-tactile mappings. The parameter

mappings for HALA that are described in the following sections were designed based on data from these prior works in an attempt to increase the perceptual congruence between data and display dimensions and therefore reduce the amount of attention that needs to be spent on understanding the feedback.

The following sections will discuss the specifics of the feedback design, with this discussion being separated into the two parameters which the system gives feedback on.

3.1.1 Position of Wrist. Understanding the angle of the wrist during repetitions allows the user to get used to how long the movement should take and also ensures that the wrist is extended and flexed to the appropriate degree. The angle is calculated using the tilt in the x-axis of the IMU. The maximum extension and flexion values (wrist fully bent to a comfortable degree in both upward and downward directions) are calibrated before exercise begins. This is both to inform the system of what points during the motion to give feedback and to ensure the user does not over-extend the wrist. During repetitions, vibrotactile cues are given when the user's wrist reaches: maximum extension, maximum flexion and the mid-point between the two. Wrist position is mapped to vibrotactile duration (Table 1) based on prior experimental work [7] which found that increasing the duration of a vibration cue is perceived as an increase in a value relating to some generalised value of stress — the specific context in this case being the stress of a tendon being flexed.

3.1.2 Speed of Movement. The time between full extension and flexion should be between 6 and 10 seconds. On each repetition, once the wrist is fully flexed and is being raised back to the starting position, a vibration cue will be given that will convey whether the just-completed flexion movement was too slow, too fast or within the ideal range. The tempo (or rate) of a repeating vibration pulse is used to convey this, due to the semantic similarity between *speed* of the flexion and tempo or *speed* of vibration. If the movement was in the correct time range (6-10 secs), a constant vibration with no pulsing and therefore with no tempo is given to indicate this. If the just-completed movement was too slow (>10 secs), a cue consisting of a pulsing vibration with a slow tempo will be given (50ms ON, 800ms OFF). Alternatively if the movement was too fast (<6 secs), a cue with a faster tempo will be given (50ms ON, 15ms OFF). Figure 3 shows a visualisation of the vibration patterns given when a movement was in range, too fast or too slow.

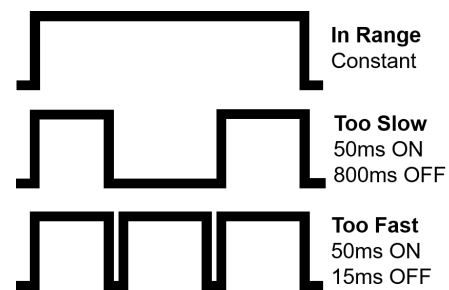


Figure 3: Visualisations of waveforms of a repeatedly pulsing vibrations used to convey the speed of movement.

4 PROPOSED EVALUATION

In order to understand the potential for the HALA system, evaluations would be necessary. The following sections will discuss how the system could be potentially evaluated from HCI and physiotherapy perspectives.

4.1 Human-Computer Factors

Prior work has shown that the data-vibration mapping used in a vibrotactile feedback system affects how the feedback is perceived [7]. Therefore, in the case of HALA, it would be necessary to not only ensure that user's can perceive and understand the feedback being presented to them, but that it is optimised to be as easy-to-understand as possible. Furthermore, the position of the actuator would need to be investigated further as it's placement may impact how the vibrations are perceived, as well as usability and comfort of the system. Another HCI aspect that could be investigated is how well users can carry out exercises using HALA while attending to some other stimulus such as a conversation, or watching television. All of these factors could be tested in a laboratory study.

4.2 Physiotherapy Factors

This system would need to be evaluated from a physiotherapy perspective as well as from an HCI perspective, to ensure that it effectively reaches the goals of the type of rehabilitation that is intended to be conducted. Georgiou *et al.*'s study of vibrotactile feedback for gait therapy [9] is a good example of how to evaluate a system from a physiotherapy perspective. In their study, physiotherapists observed the patient using the system and the researchers conducted a *post-hoc* interview to establish the physiotherapists positive and negative reactions to the system. Furthermore, this method gives the physiotherapists an opportunity to assess how the system may be helping or hindering the patient's form and timing etc.

5 DISCUSSION AND FURTHER RESEARCH

The prototype was developed in conjunction with a physiotherapist and an adviser who is clinically diagnosed with LE, to evaluate that the hardware would not interfere with or constrict any muscles involved in the exercise and the vibrations used would not have any similar adverse effect. Furthermore, the vibration parameters used were evaluated early in the design during informal trials conducted at the Scottish Information and Computing Science Alliance's Demofest. From this initial design and development, a clear progression from is to further evaluate the system with a larger population of potential users who are diagnosed with tennis elbow, investigating if the system can be an effective long-term intervention to improve adherence to home based exercise programs. Finally, as the goal of this system is to allow the user to attend to their exercises peripherally, a secondary, yet equally important, aspect of this investigation must centre around measuring the user's attention to a task while carrying out exercises. Beyond this straightforward further development, there are many potential applications which researching systems such as HALA in the future may contribute toward, including:

5.1 Applications to Other Areas of Physiotherapy

The barriers to adherence which this system aims to combat are problematic not only to LE rehabilitation, but to the entire spectrum of conditions that require physiotherapeutic rehabilitation. Therefore, there is potential for future research into application of this system, or similar systems, as an aid for rehabilitation to a number of other orthopaedic or neurological conditions where home based exercise is required.

5.2 Physiotherapy Post COVID-19

The introduction to this paper discussed the challenges that are faced by physiotherapy services due to the COVID-19 pandemic. However, some are already looking at the lessons being learned from the situation and advocating for continued funding and research into telerehabilitation and other at-home physiotherapy interventions [21, 24]. This form of treatment has proven to be more than a stop-gap until the pandemic has been resolved, with research in the past year indicating that at-home physiotherapy has been effective for post-operation care for procedures such as esophageal cancer [28] and hip replacements [20]. Therefore, it seems that telerehabilitation will remain in some form post-COVID, and as such, now is an ideal time to conduct further research into haptics and other non-visual based feedback systems such as HALA as physiotherapy interventions within the HCI community.

5.3 Integration with Pre-Existing Hardware

Another area for future work in this area, is the integration of feedback systems such as HALA into pre-existing technologies such as smartphones and smartwatches. The HALA prototype presented in this paper runs using a simple Unity application which has the potential to be ported to Android or iOS, making it accessible to these mobile devices. Furthermore, mobile devices such as the iPhone and Apple Watch are including more sophisticated haptic feedback actuators like Apple's Taptic Engine [31], allowing for expressive feedback as is used in the HALA system. Further researching the potential for using smartphone and smartwatch devices for feedback during at-home physiotherapy is an important next step, as using these devices have the advantages that the user already owns them and is familiar with their usage, reducing the financial cost to the user and/or healthcare provider.

5.4 Potential for Gamification

Gamification, "the use of video game elements (rather than full-fledged games) to improve user experience and user engagement in non-game services and applications" [6], has been used by researchers in the HCI community as an approach for developing successful physiotherapy interventions and tools for several years [8, 10, 13, 19] and physiotherapy has been shown via a systematic review of gamification literature to be one the most common applications of gamification [15]. Furthermore, gamified systems have been shown to aide in encouraging positive behavioural changes [18]. Therefore, digital interventions such as HALA are inherently applicable to gamification, since they already communicate with a computer system and have the potential to be integrated into a gamification system. This would allow patients to reap the potential

benefits that this technique can offer, including encouraging positive behaviours such as adherence to prescribed physiotherapy programs.

5.5 Implications for Tactile Feedback Design

Research into this system presents an opportunity to further understand the perception of parameter mappings for tactile displays. Previous work has investigated using psychophysical methods to evaluate, in a non-applied setting, the perceptual congruence between a tactile parameter and data parameter [7], however these mappings have not yet been evaluated in an applied context. Therefore systems like HALA can provide a testbed for evaluating more low-level parameter mapping research in an applied setting.

6 CONCLUSIONS

Lateral epicondylitis (LE) or tennis elbow is a highly prevalent musculoskeletal condition that can seriously impact the wellbeing of individuals that are suffering [1, 14] and causes significant economic impact as a result [11]. Poor adherence to prescribed at-home physiotherapy programs is a common reason for poor outcomes from physiotherapy. Common barriers to adherence are: low self-efficacy [12], patients' belief that they would be more willing to carry out exercises if they were accompanied by a friend or partner [5], and inconvenience or conflict with everyday life [5]. Furthermore, the strain on health services and necessity for reduction in elective treatment caused by the COVID-19 pandemic has meant that access to physiotherapy has been restricted and treatment has moved to virtual and at-home treatment [27].

In response to these barriers to physiotherapy access and adherence, we present HALA (Haptically Augmented Lateral Elbow Tendinopathy Aid), a prototype system which uses vibration feedback to convey information to the user regarding the speed of movement and position of the wrist gathered from a sensor worn on the hand during eccentric exercise of the wrist extensors, a commonly prescribed exercise for LE rehabilitation [4] (Figure 1). The goal of this vibrotactile approach is to allow the user to attend to attend to some other task concurrently with exercise, such as holding a conversation or watching television, therefore confronting the barriers to adherence of: being able to be accompanied by a friend or partner during physiotherapy [5] and fitting physiotherapy around everyday activities [5] such as conversations or watching television.

Finally, this paper discussed areas of future work for this prototype and similar approaches, including applications to other areas of physiotherapy, confronting the challenges faced by physiotherapists in the wake of the COVID-19 pandemic and implication for tactile feedback design.

Future research into systems such as the HALA prototype are integral to reduce the risk of abandonment of prescribed physiotherapy programs for LE and other conditions and therefore increase the chances of successful recovery. Furthermore, the increase in at-home physiotherapy solutions and reduction in elective treatments made necessary by the COVID-19 pandemic mean that further research from the HCI community on developing tools that can support patients and physiotherapists will be key to their success. We hope that the prototype we have proposed here can serve as a stimulus for discussion in the community about how we can utilise

our expertise in non-visual feedback methods such as vibration to aide physiotherapists in reducing patients' barriers to adherence.

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